Security incident report

Section 1: Identify the network protocol involved in the incident
The main network protocol involved in this incident was HTTP (Hypertext Transfer Protocol). HTTP was used by the browser to request webpages from yummyrecipesforme.com and the fake website, greatrecipesforme.com. DNS (Domain Name System) was also used to find the IP addresses of both websites before the browser connected to them. Both HTTP and DNS operate at the application layer of the TCP/IP model.

Section 2: Document the incident
The website yummyrecipesforme.com experienced a security incident caused by a former employee who gained unauthorized access to the website's admin panel through a brute force attack. The attacker repeatedly tried default passwords until they successfully logged in. They then changed the website's source code by adding malicious JavaScript that prompted visitors to download an executable file. After running the file, users were redirected to a fake website called greatrecipesforme.com, which contained malware. The incident was discovered after several customers contacted the helpdesk to report unexpected download requests, website redirection, and slow computer performance. During the investigation, the cybersecurity team used tcpdump in a sandbox environment to monitor network traffic. The traffic showed DNS requests and HTTP connections to both websites. Further investigation confirmed that the attacker had changed the admin password and that the original password was still set to its default value before the attack.The main cause of the incident was weak password security and the absence of protection against repeated login attempts.

Section 3: Recommend one remediation for brute force attacks
I recommend limiting the number of failed login attempts on the website's admin panel. For example, after five unsuccessful login attempts, the system should temporarily lock the account or delay further login attempts.

This will make it harder for attackers to keep guessing passwords until they find the correct one. The security team should also monitor failed login attempts to identify suspicious activities. Implementing this measure will help reduce the risk of future brute force attacks and protect the website from unauthorized access.